Technology Stack

Date	15 March 2026
Team ID	SWTID-2026-5240
Project Name	Streamlining IT Procurement: Automating Standard Laptop Procurement
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Web pages-ah fast-a render panna, responsive layout grid configure panna, matrum preview UI component-ah dynamic-ah display panna use aagudhu.
2	Backend / Server-Side	Python (Flask Framework)	Lightweight framework, Computer Vision libraries (OpenCV) oda seamlessly integrate aagum, fast execution matrum easy routing-kku use aagudhu.
3	Database / Data Storage	SQLite / Local File System	Upload panna wireframe images, generated .html/.css/.js files, matrum extracted JSON layout metadata-va store panna use aagudhu.
4	Cloud / Hosting / Deployment	Render / Localhost (VS Code Server)	Easy deployment, cost-effective hosting, matrum Python Flask web applications-ah easily run panna use aagudhu.
5	Version Control & CI/CD	Git & GitHub	Source code files-ah manage panna, project version control maintain panna, matrum team collaboration-kku use aagudhu.
6	Third-Party APIs / Other Tools	OpenCV & Tesseract OCR / EasyOCR	Sketch images-la irundhu UI bounding box contours detect panna, component classification grid parsing-kku, matrum text labels extract panna use aagudhu.